

Ligjerata 10d

Bymimi termik vëllimor i trupave

$$\Delta V_r = \frac{\Delta V}{V_0} \quad \Delta V = V - V_0$$

$$\beta = \frac{\Delta V_r}{\Delta t} \quad \beta = \frac{1}{\Delta t} \cdot \frac{\Delta V}{V_0}$$

$$\beta = \frac{1}{\Delta t} \frac{V - V_0}{V_0} \cdot V_0 \Delta t \Rightarrow \beta \cdot V_0 \cdot \Delta t = V - V_0$$

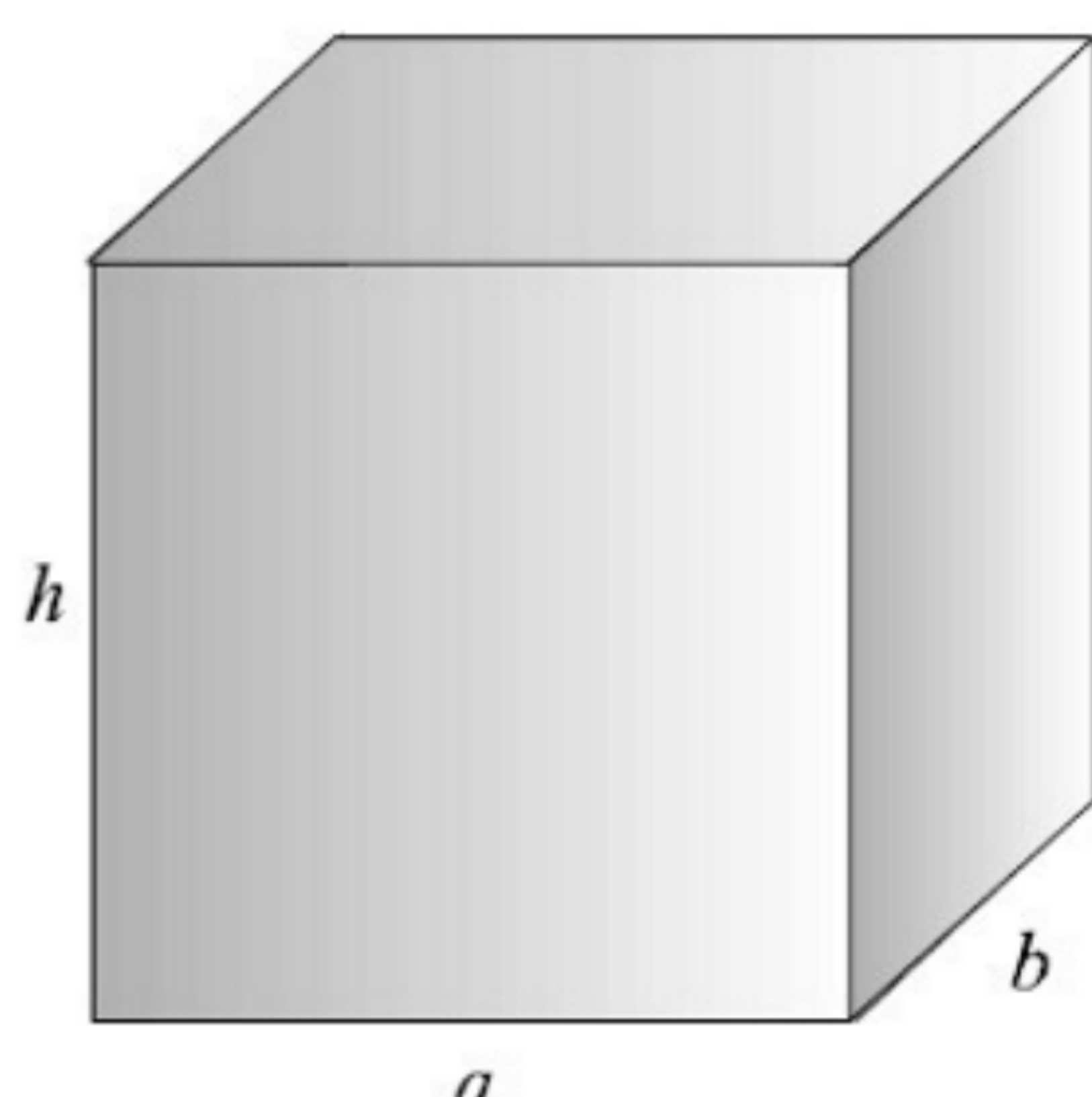
$$\beta V_0 \cdot \Delta t + V_0 = V$$

$$V = V_0(1 + \beta \Delta t)$$

ose

$$V = V_0(1 + \beta t)$$

$$\boxed{\beta = 3\alpha}$$



$$a = a_0(1 + \alpha t)$$

$$b = b_0(1 + \alpha t)$$

$$h = h_0(1 + \alpha t)$$

$$V = a \cdot b \cdot h$$

$$= a_0(1 + \alpha t) \cdot b_0(1 + \alpha t) \cdot h_0(1 + \alpha t)$$

$$= a_0 b_0 h_0 (1 + \alpha t)^3$$

$$V_0 = a_0 b_0 h_0$$

$$V = V_0 (1 + 3\alpha t + 3\alpha^2 t^2 + \alpha^3 t^3)$$

$$V = V_0(1 + 3\alpha t) ; \beta = 3\alpha$$

$$V = V_0(1 + \beta t)$$

$$\Delta l = l_2 - l_1$$

$$\Delta t = t_2 - t_1$$

$$\alpha = \frac{1}{\Delta t} \cdot \frac{\Delta l}{l_0} = \frac{1}{\Delta t} \frac{l_2 - l_1}{l_1} \rightarrow \alpha \cdot \Delta t \cdot l_1 = l_2 - l_1$$

$$l_2 = l_1 + l_1 \alpha \Delta t$$

$$l_2 = l_1(1 + \alpha \Delta t)$$

$$\beta = \frac{1}{\Delta t} \frac{\Delta V}{V_0} = \frac{1}{\Delta t} \frac{V_2 - V_1}{V_1} \rightarrow \beta \Delta t \cdot V_1 = V_2 - V_1$$

$$V_2 = V_1 + V_1 \beta \Delta t$$

$$\boxed{V_2 = V_1(1 + \beta \Delta t)}$$

Vartësia e densitetit të trupave nga temperatura

$$t_0 = 0^\circ \text{C}$$

$$m_0 = \rho_0 V_0$$

$$t \neq 0^\circ \text{C}$$

$$m_t = \rho(t) V_t$$

$$m_0 = m_t$$

$$\rho(t) = \frac{\rho_0 V_0}{V_t}$$

$$\rho(t) = \frac{\rho_0 V_0}{V_0(1 + \beta t)}$$

$$\boxed{\rho(t) = \frac{\rho_0}{1 + \beta t}}$$

$$\rho_0 V_0 = \rho(t) V_t / : V_t$$

$$V_t = V_0(1 + \beta t)$$